project-1

2025-09-26

Task 1

I am going to knit directly to a pdf file using the tinytex library.

A .Rmd file contains the R source code and text that is used to generate the .pdf file. The .pdf file is usually considered the final product.

```
medicaldata <- read.csv("medicaldata1.csv", header = TRUE)
head(medicaldata, n = 6)
```

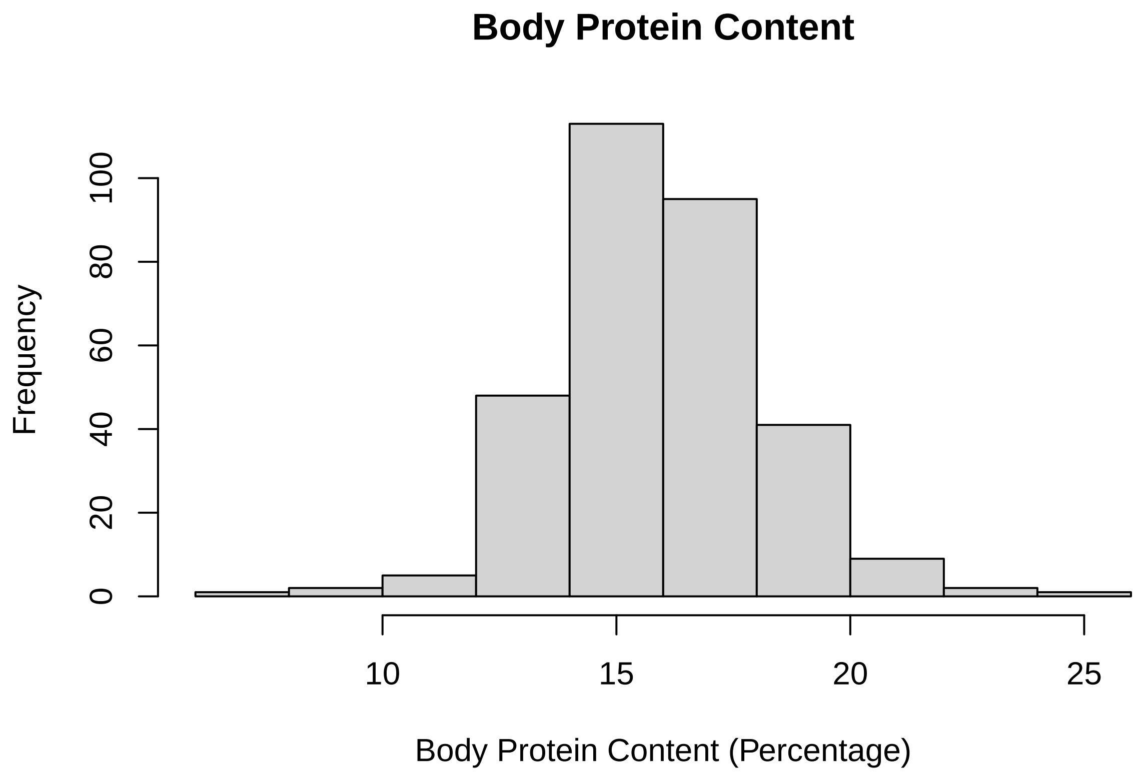
```
##   Age Height Weight Gallstone_Status Comorbidity CAD Diabetes_Mellitus
## 1  57    170  143.5                0           0   0                 0
## 2  53    150  108.4                1           1   0                 1
## 3  54    155   80.2                1           0   0                 0
## 4  63    150   69.0                1           0   0                 0
## 5  73    160  104.6                1           1   0                 1
## 6  56    158  100.5                0           0   0                 0
##   Total_Body_Water Total_Body_Fat_Ratio Lean_Mass Body_Protein_Content
## 1             66.2                42.30     57.70                 7.99
## 2             40.0                50.92     48.99                 9.05
## 3             32.8                46.13     53.87                 9.63
## 4             29.9                42.46     57.39                10.49
## 5             52.7                34.20     65.77                11.31
## 6             35.9                49.80     50.25                11.41
##   Hepatic_Fat_Accumulation Glucose Triglyceride
## 1                        3     118          112
## 2                        2     119           97
## 3                        2     111          104
## 4                        2     106           96
## 5                        3     230          208
## 6                        3     129           81
```

Task 2

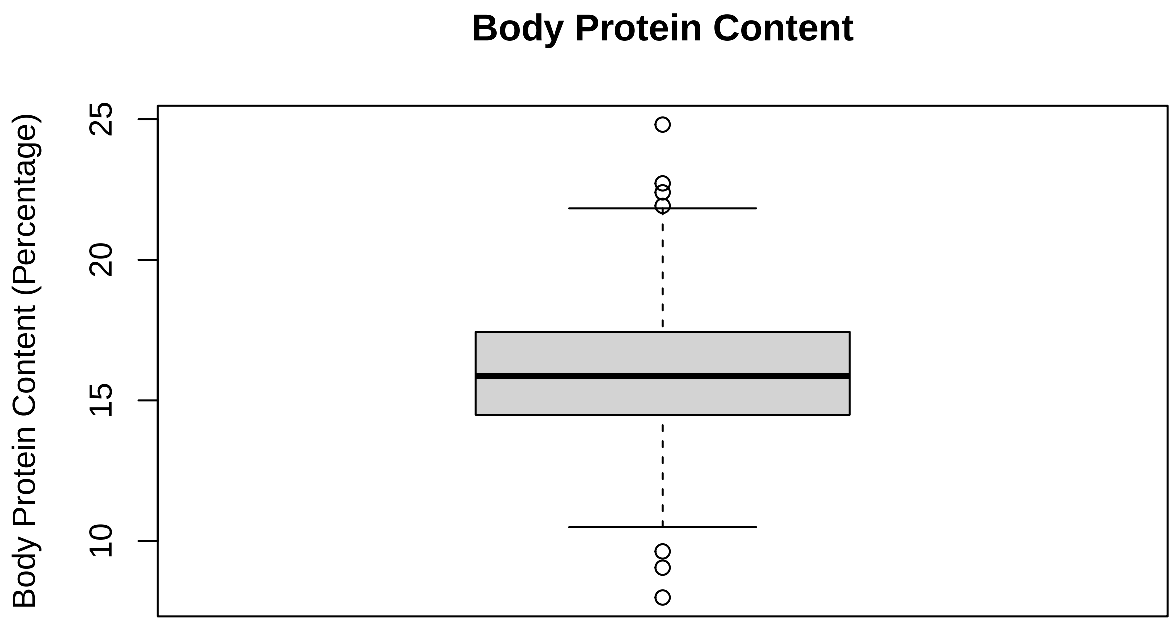
```
medicaldata$Gallstone_Status <- as.factor(medicaldata$Gallstone_Status)
medicaldata$Comorbidity <- factor(medicaldata$Comorbidity, ordered = TRUE, levels = c(0, 1, 2, 3))
medicaldata$CAD <- as.factor(medicaldata$CAD)
medicaldata$Diabetes_Mellitus <- as.factor(medicaldata$Diabetes_Mellitus)
medicaldata$Hepatic_Fat_Accumulation <- factor(medicaldata$Hepatic_Fat_Accumulation, ordered = TRUE, le
```

Task 3

```
hist(medicaldata$Body_Protein_Content, main = "Body Protein Content", xlab = "Body Protein Content (Perc
```



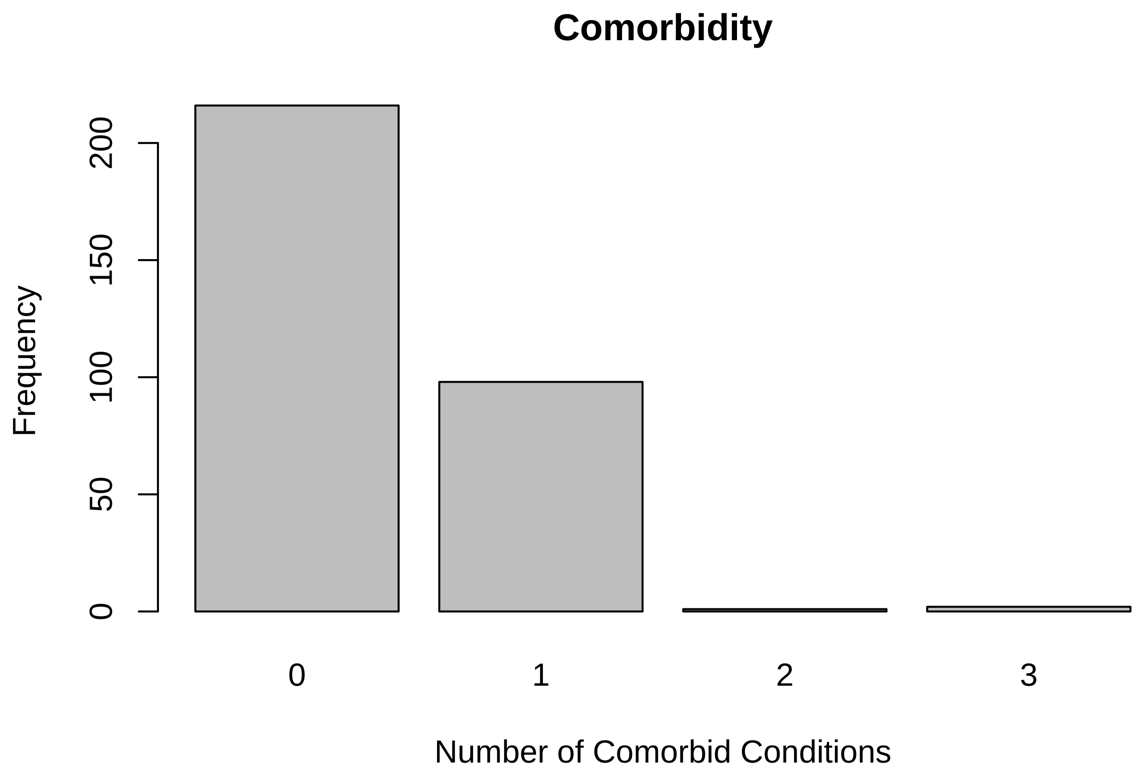

```
boxplot(medicaldata$Body_Protein_Content, main = "Body Protein Content", ylab = "Body Protein Content (P
```



Based on the above boxplot and histogram, the body protein content appears to be normally distributed. The boxplot shows only 7 outliers out of a dataset of 317 individuals. The histogram is shaped like a bell curve, implying normality. The mean body protein content is 15.97095% with a standard deviation of 2.267927%.

Task 4

```
comorbidity_frequency <- table(medicaldata$Comorbidity)
barplot(comorbidity_frequency, main = "Comorbidity", xlab = "Number of Comorbid Conditions", ylab = "Fr
```

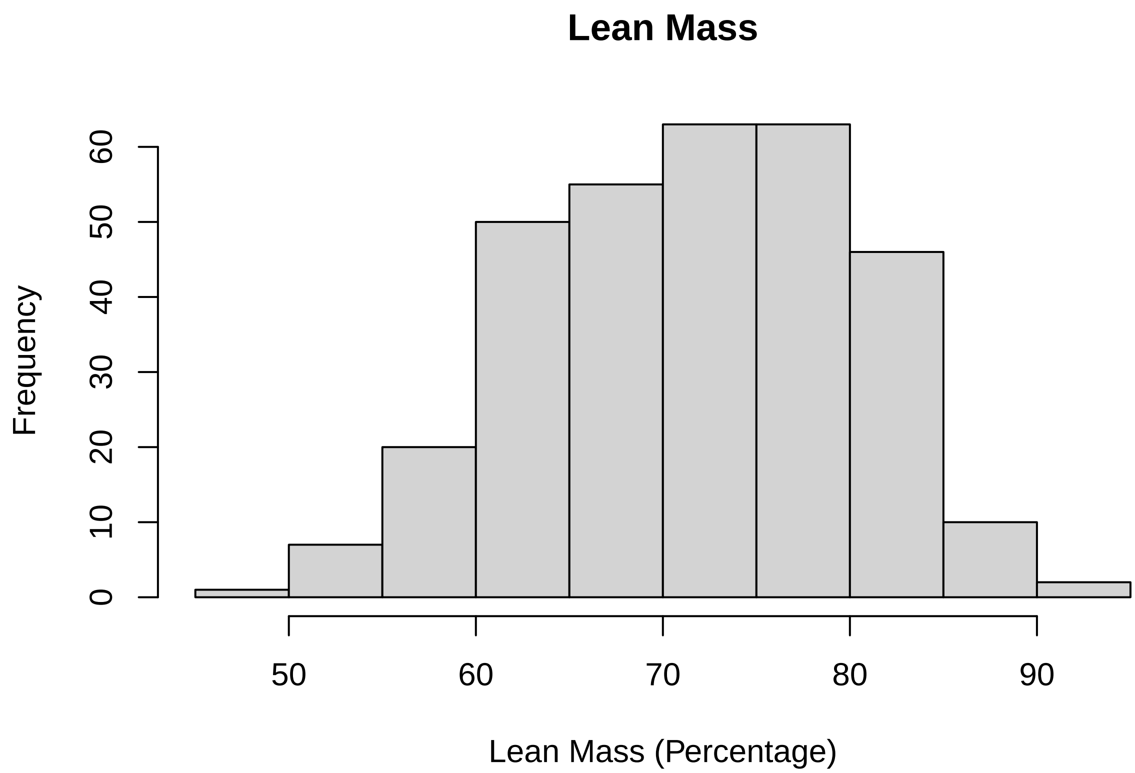


The graph of the number of comorbid conditions present in the indiduals in the medical data set is skewed right. The most frequent number of comorbid conditions present is zero, which occurs 216 times. 98 individuals have one comorbid condition. 1 individual has two comorbid conditions. 2 individuals have three comorbid conditions.

Task 5

Task 6

```
library(moments)
hist(medicaldata$Lean_Mass, main = "Lean Mass", xlab = "Lean Mass (Percentage)")
```




```r
skewness(medicaldata$Lean_Mass)
```

```
## [1] -0.1179956
```

```r
kurtosis(medicaldata$Lean_Mass)
```

```
## [1] 2.497133
```

```r
shapiro.test(medicaldata$Lean_Mass)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  medicaldata$Lean_Mass
## W = 0.99178, p-value = 0.07566
```

The skewness of the Lean Mass histogram is -0.1179956 and the kurtosis is 2.497133. Running the Shapiro-Wilk normality test produces values of W = 0.99178 and a p-value = 0.07566. The null hypothesis is that the Lean Mass variable is normally distributed. The alternative hypothesis is that the Lean Mass variable is not normally distributed. Using a significance level of 3%, we can make the decision to not reject the null hypothesis becaues the p-value = 0.07566 > 0.03. We can conclude that there is enough evidence that the normality of the Lean Mass variable seems plausible. ## **Task 7**

Task 8

Task 9